

Shravani Mahadeshwar

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Education

09.2025 – 09.2027 (expected)
M.Sc. Applied Data Science — **Frankfurt, Germany**
Frankfurt School of Finance and Management
Grade: 1.9 (German scale) | Coursework: Machine Learning, Deep Learning, AI Ethics, Database and Cloud

09.2021 – 03.2024
B.Sc. Data Science and Business Analytics — **Mumbai, India**
University of Mumbai
Grade: 9.52 / 10.00 | Coursework: NLP, Computer Vision, Pattern Recognition, Time Series Analysis

Skills

Programming	Python, SQL, R, Git
Machine Learning	Explainability (SHAP), Mechanistic Interpretability, LLM Evaluation, RAG Pipelines
Automation & Cloud	n8n, Power Automate, Selenium, REST APIs, AWS (EC2, S3), Terraform
Data Visualization	Power BI, Tableau, Dashboard Design
Microsoft 365	Excel (Advanced), PowerPoint, SharePoint, Copilot
Languages	English (C1), German (B1)

Professional Experience

06.2024 – 01.2025
Junior Data Science and Business Analyst — **Mumbai, India**
JM Financial Services Limited
- Designed and maintained 4+ enterprise Power BI dashboards by extracting, transforming, and integrating data using SQL from multiple financial databases, enabling executive KPI reporting and client yield analysis.
- Architected a production-ready OCR and signature validation pipeline using OpenCV and EasyOCR with Flask APIs and Dockerized deployment, reducing KYC manual screening by 60%.
- Engineered a Python-Outlook automation framework for recipient-specific report distribution with validation and tracking logic, cutting reporting turnaround time by 70%.
- Queried and analyzed customer data using SQL, applying RFM analysis, feature engineering, and K-Means clustering to deliver data-driven customer segmentation and actionable recommendations.

09.2023 – 02.2024
Data Science Intern — **Mumbai, India**
JM Financial Services Limited
- Built a Selenium-based RPA system for FTP-controlled mutual fund report processing, reducing manual intervention by 90%.
- Developed an NLP pipeline on NIFTY 500 financial news using VADER sentiment analysis to transform unstructured text into structured investment signals.

Projects

03.2026
LLM Audit Bench
- Benchmarked 5 open-source LLMs across trustworthiness dimensions using CrowS-Pairs bias evaluation, perturbation-based robustness testing, SHAP attribution analysis, and privacy-risk assessment.
- Constructed a composite trustworthiness metric, enabling ranked model selection for deployment suitability.

02.2026
Mechanistic Analysis of Transformer Hallucinations
- Investigated factual hallucinations in GPT-Neo-125M through mechanistic interpretability techniques, tracing token propagation across attention heads and residual pathways.
- Developed attribution heatmaps, attention-flow visualisations, and 3D circuit graphs to analyse how erroneous generations emerge within transformer architectures.

10.2025 – 11.2025
Decoding ESG: Measuring Clarity and Greenwashing in Corporate Disclosures
- Conducted a comparative market benchmarking study across 10 years of corporate disclosures from S&P 50 companies, structuring unstructured data into a Python pipeline to support cross-company analysis.
- Designed quantitative benchmarking metrics, including a Greenwashing Risk Index and Concreteness Score, to rank companies on disclosure quality and inform data-driven market intelligence reporting.

Certificates

07.2026 – 08.2026 BlueDot Impact: Technical AI Safety Fellowship

05.2025 – 07.2025 McKinsey Forward Program